

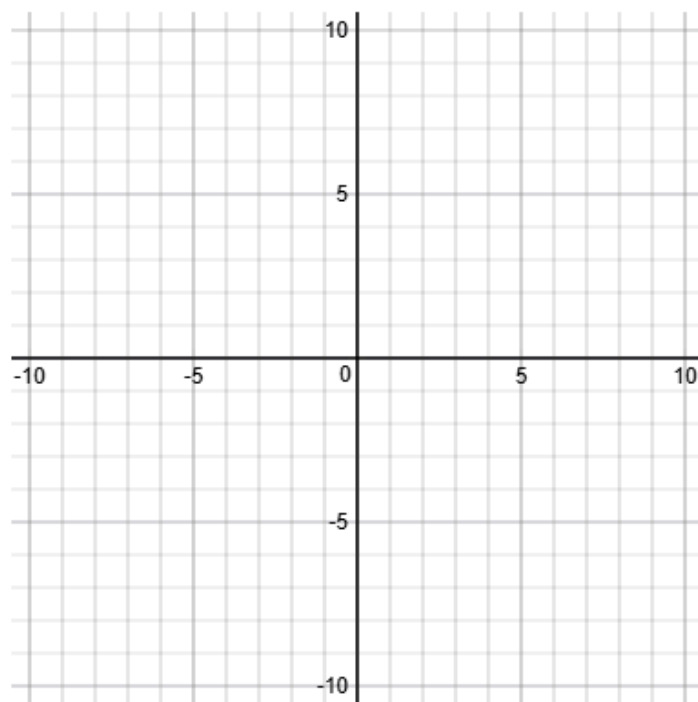
Calculus I Lesson 02

1) $f(x) = \frac{\cos x}{x - \pi}$

a) Graph $f(x) = \frac{\cos x}{x - \pi}$.

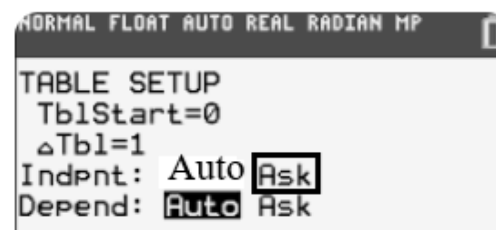
Vertical Asymptote is:

$x = \underline{\quad ? \quad}$



Using TI-84: Graphing Function:
Press [Y=]
Input: $Y1 = \cos(X)/(X - \pi)$
Press [GRAPH]

Finding Function Values:
Press [2nd] [WINDOW]
Set Indpnt on Ask



Press [2nd] [GRAPH]

Input x-values: $\pi - 0.5$, $\pi - 0.1$, $\pi - 0.0001$,...

b) Complete the following table.

x	$f(x)$
$\pi - 0.5$	
$\pi - 0.1$	
$\pi - 0.0001$	
$\pi - 0.0000001$	

c) Complete the following table.

x	$f(x)$
$\pi + 0.5$	
$\pi + 0.1$	
$\pi + 0.0001$	
$\pi + 0.0000001$	

d) $\lim_{x \rightarrow \pi^-} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

limit of $f(x)$ as x approaches π from the left

$\lim_{x \rightarrow \pi^+} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

limit of $f(x)$ as x approaches π from the right

$\lim_{x \rightarrow \pi} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

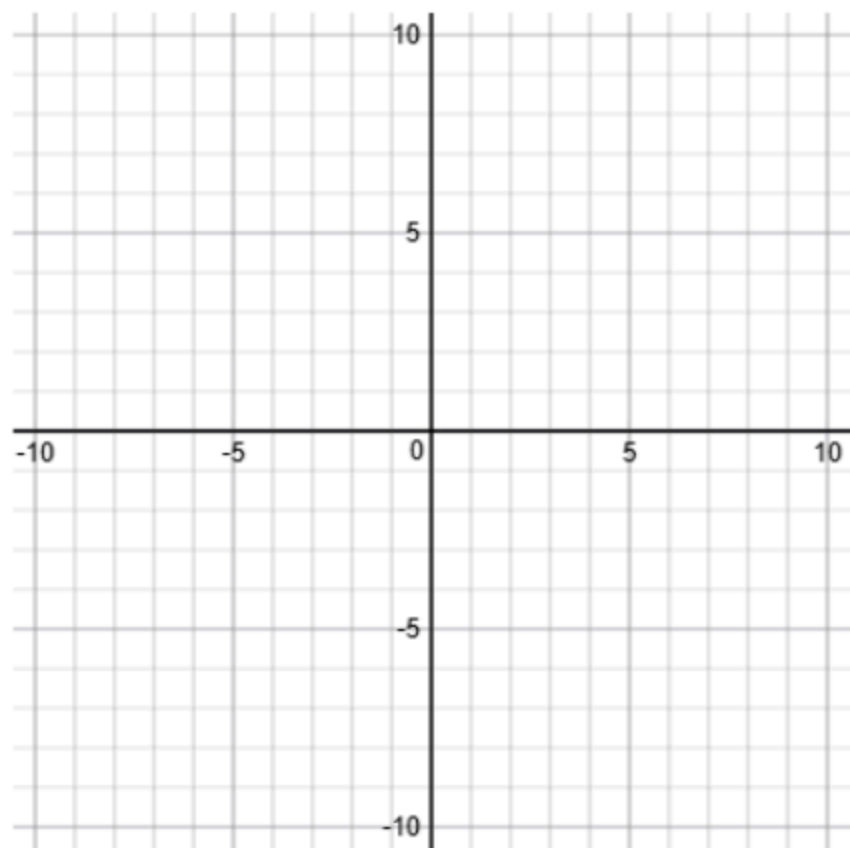
limit of $f(x)$ as x approaches π

2) $f(x) = \cot x$

a) Graph $f(x) = \cot x$ $0 \leq x \leq 2\pi$

Vertical Asymptotes are:

$x = \underline{\quad ? \quad}$ and $\underline{\quad ? \quad}$



b) Complete the following table.

x	$f(x)$
$\pi - 0.5$	
$\pi - 0.1$	
$\pi - 0.0001$	
$\pi - 0.0000001$	

c) Complete the following table.

x	$f(x)$
$\pi + 0.5$	
$\pi + 0.1$	
$\pi + 0.0001$	
$\pi + 0.0000001$	

d) $\lim_{x \rightarrow \pi^-} f(x) = \underline{\hspace{1cm}} ?$

limit of $f(x)$ as x approaches π from the left

$\lim_{x \rightarrow \pi^+} f(x) = \underline{\hspace{1cm}} ?$

limit of $f(x)$ as x approaches π from the right

$\lim_{x \rightarrow \pi} f(x) = \underline{\hspace{1cm}} ?$

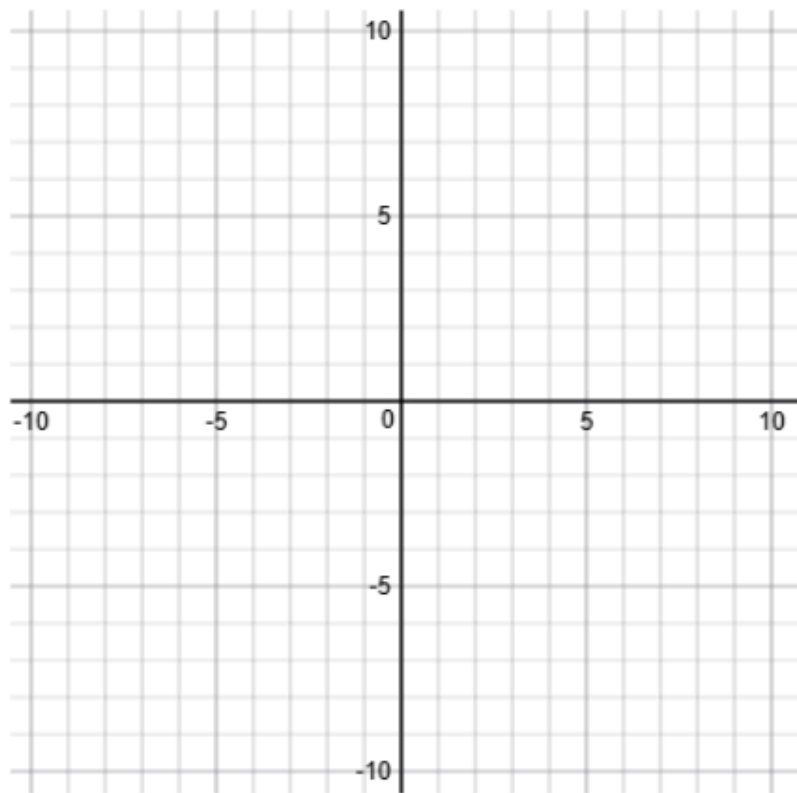
limit of $f(x)$ as x approaches π

$$3) f(x) = \frac{x}{\sin x - \pi/2}$$

$$\text{a) Graph } f(x) = \frac{x}{\sin(x - \pi/2)} \quad 0 \leq x \leq 2\pi$$

Vertical Asymptotes are:

$x = \underline{\quad ? \quad}$ and $x = \underline{\quad ? \quad}$



b) Complete the following table.

x	$f(x)$
$\pi/2 - 0.5$	
$\pi/2 - 0.1$	
$\pi/2 - 0.0001$	
$\pi/2 - 0.0000001$	

c) Complete the following table.

x	$f(x)$
$\pi/2 + 0.5$	
$\pi/2 + 0.1$	
$\pi/2 + 0.0001$	
$\pi/2 + 0.0000001$	

d) $\lim_{x \rightarrow \pi/2^-} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

limit of $f(x)$ as x approaches $\pi/2$ from the left

$\lim_{x \rightarrow \pi/2^+} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

limit of $f(x)$ as x approaches $\pi/2$ from the right

$\lim_{x \rightarrow \pi/2} f(x) = \underline{\hspace{1cm}} ? \hspace{1cm}$

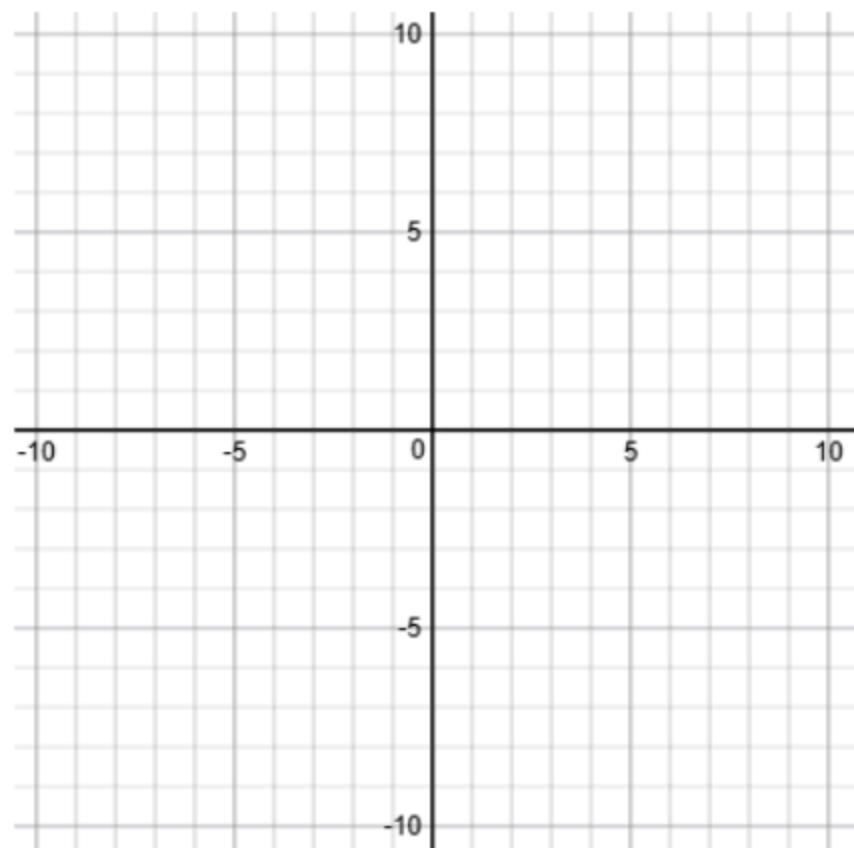
limit of $f(x)$ as x approaches $\pi/2$

4) $f(x) = \tan x$

a) Graph $f(x) = \tan x$ $0 \leq x \leq 2\pi$

Vertical Asymptotes are:

$x = \underline{\quad ? \quad}$ and $x = \underline{\quad ? \quad}$



b) Complete the following table.

x	$f(x)$
$3\pi/2 - 0.5$	
$3\pi/2 - 0.1$	
$3\pi/2 - 0.0001$	
$3\pi/2 - 0.0000001$	

c) Complete the following table.

x	$f(x)$
$3\pi/2 + 0.5$	
$3\pi/2 + 0.1$	
$3\pi/2 + 0.0001$	
$3\pi/2 + 0.0000001$	

d) $\lim_{x \rightarrow 3\pi/2^-} f(x) = \underline{\hspace{1cm}} ?$

$\lim_{x \rightarrow 3\pi/2^+} f(x) = \underline{\hspace{1cm}} ?$

$\lim_{x \rightarrow 3\pi/2} f(x) = \underline{\hspace{1cm}} ?$

limit of $f(x)$ as x approaches $3\pi/2$ from the left

limit of $f(x)$ as x approaches $3\pi/2$ from the right

limit of $f(x)$ as x approaches $3\pi/2$

$$5) f(x) = \frac{x}{\tan(x - \pi/3)}$$

$$a) \text{ Graph } f(x) = \frac{x}{\tan(x - \pi/3)} \quad 0 \leq x \leq 2\pi$$

$$\text{Note: } \tan(0) = 0; \quad \tan(\pi) = 0; \quad \tan(2\pi) = 0$$

$$\text{Hence, } f(x) = \frac{x}{\tan(x - \pi/3)} \text{ is undefined when } x - \pi/3 = 0, \pi, 2\pi$$

$$\text{Meaning, } f(x) = \frac{x}{\tan(x - \pi/3)} \text{ is undefined when } x = \pi/3, \pi + \pi/3, 2\pi + \pi/3$$

Vertical Asymptotes are:

$$x = \underline{\quad ? \quad} \text{ and } x = \underline{\quad ? \quad}$$

b) Complete the following table.

x	$f(x)$
$4\pi/3 - 0.5$	
$4\pi/3 - 0.1$	
$4\pi/3 - 0.0001$	
$4\pi/3 - 0.0000001$	

c) Complete the following table.

x	$f(x)$
$4\pi/3 + 0.5$	
$4\pi/3 + 0.1$	
$4\pi/3 + 0.0001$	
$4\pi/3 + 0.0000001$	

d) $\lim_{x \rightarrow 4\pi/3^-} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches $4\pi/3$ from the left

$\lim_{x \rightarrow 4\pi/3^+} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches $4\pi/3$ from the right

$\lim_{x \rightarrow 4\pi/3} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches $4\pi/3$

e) $\lim_{x \rightarrow \pi/3^-} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches $\pi/3$ from the left

$\lim_{x \rightarrow \pi/3^+} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches $\pi/3$ from the right

$\lim_{x \rightarrow \pi/3} f(x) = \underline{\quad ? \quad}$

limit of $f(x)$ as x approaches $\pi/3$

